Sir Syed University of Engineering & Technology

Department of Computer Science & Information Technology

CS-467- Software Project Management

Lecture No. 6

SOFTWARE PROJECT RISK MANAGEMENT

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List of Topic to be Discuss in this lecture

- What is Risk?
- What is Software Risk
- Software Risk Management Processes
- Software Risk Identification
- Risk Analysis
  - Perform Qualitative Risk Analysis
  - Perform Quantitative Risk Analysis
- Plan Risk Response
- Software Risk Monitoring

Knowledge Area -Risk Management

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Risk Management		Plan Risk Management Identify Risk Perform Qualitative Risk Analysis Perform Quantitative Risk Analysis Plan Risk Response		Monitor & Control Risk	

What is Risk?

- **Risk** is an expectation of loss, a potential problem that may or may not occur in the future. It is generally caused due to lack of information, control or time.

- **Risk** is an uncertain event that could have a positive or negative outcome.

- **Risk** is everywhere: whether you're driving, sky diving, crossing the street, trading stocks, or swimming with sharks. When you do something risky, you must calculate whether the potential reward is worth the potential risk. Some things are easy — the reward of driving from Point A to Point B is usually considered worth the risk of a traffic accident. For every risk there is some reward. If you're lucky, the reward works in your favor, like buying low and selling high in the stock market.

What is Software Risk?

A possibility of suffering from loss in software development process is called a **software risk**.

Loss can be anything, increase in production cost, development of poor quality software, not being able to complete the project on time. Software risk exists because the future is uncertain and there are many known and unknown things that cannot be incorporated in the project plan.

A software risk can be of two types

1. **Internal / Business risks** that are within the control of the project manager
2. **External / Pure risk** that are beyond the control of project manager

Software Risk Management Processes

- Software Risk Identification – **what are the risks to a project?**
- Software Risk Analysis - **which ones are really serious?**
- Plan Risk Response - **what shall we do?**
- Software Risk Monitoring – **has the planning worked?**

Threats and Opportunities: Negative Risks and Positive Risks

- **Risk** is an uncertain event which may have a positive or negative effect on the project if it occurs. It can be in the form of **opportunities** (Positive Risks) as well as **threats** (Negative Risks).

- **Threats:** A Threat is a risk with a harmful effect on the project .e.g an untrained workforce can have can create a faulty product which can cost the project more time and money. Lack of a proper communication system in the organization can result in a decreased productivity.

- **Opportunities:** An Opportunity is a risk with a positive effect on the project. By hiring a professional and trained resource such as an expert software programmer, you could end up saving time by getting work done 30% faster.

- **Uncertainty:** Uncertainty on the project is caused by **Lack of Information**.

Types of Business Risks

There are different types of business risks which can affect a software project:

1. **Technology risks:** Risks that assume from the software or hardware technologies that are used to develop the system.
2. **People risks:** Risks that are connected with the person in the development team.
3. **Organizational risks:** Risks that assume from the organizational environment where the software is being developed.
4. **Tools risks:** Risks that assume from the software tools and other support software used to create the system.
5. **Requirement risks:** Risks that assume from the changes to the customer requirement and the process of managing the requirements change.
6. **Estimation risks:** Risks that assume from the management estimates of the resources required to build the system

SOFTWARE RISK IDENTIFICATION

- In order to identify the risks that your project it is important to first study the problems faced by previous projects.

- Study the project plan properly and check for all the possible areas that are vulnerable to some or the other type of risks.

- It is important to conduct few brainstorming sessions to identify the known unknowns that can affect the project. Any decision taken related to technical, operational, political, legal, social, internal or external factors should be evaluated properly.

List of Boehm's Top Ten Risk Items

1. **Personnel shortfalls**
2. **Unrealistic schedules and budgets**
3. **Developing wrong software functions**
4. **Developing wrong interface**
5. **Unnecessary perfectionism**
6. **Continuous requirements change**
7. **Shortfalls in outsourced tasks**
8. **Shortfalls in outsourced components**
9. **Real-time performance problem**
10. **Development technically too difficult**

SOFTWARE RISK ANALYSIS

Risk analysis is the process of prioritizing risks based on the *probability* of the risk occurring and the *impact* it would have on the project.

There are two primary methods of risk analysis you can use on your project

1. Qualitative Risk Analysis
2. Quantitative Risk Analysis

The main difference between these two methods of risk analysis is that **qualitative risk analysis** uses a relative or descriptive scale to measure the probability of occurrence whereas **quantitative risk analysis** uses a numerical scale.

Perform Qualitative Risk Analysis

Once you've got a list of risks, you'll need to get a good idea of the probability and impact of each risk.

- Risk Exposure is comprised of Risk Impact and Probability that the risk will materialize.
- The risk impact is the cost to the project if the risk materializes. The probability is the likelihood that it will materialize.
- **Risk Exposure** = Risk Impact X Probability.

Perform Qualitative Risk Analysis

Table — Sample Qualitative Risk Impact Matrix

Risk	*Probability*	*Impact*	*Risk Exposure*
Server crashes	Low	Medium	Low
Lack of developers	High	Medium	High
Firmware changes	Low	Medium	Low
Requirement to install service packs	High	Medium	Medium
Meteorites strike company headquarters	Low	Low	Low

Perform Quantitative Risk Analysis

Expected monetary value (EMV) is a risk management technique to help quantify and compare risks in many aspects of the project. EMV is **a quantitative risk analysis technique** since it relies on specific numbers and quantities to perform the calculations, rather than high-level approximations like high, medium and low.

EMV relies on two basic numbers

Probability = P = the probability that the risk will occur

Impact = I = the impact to project if the risk occurs.

Risk Exposure = EVM = Probability X Impact

Perform Quantitative Risk Analysis

In this risk analysis example, we will use the Expected Monetary Value (EVM) technique to calculate the project risk exposure and the amount of Contingency Reserve.

Table	Sample Quantitative Risk Impact Matrix		
Risk	Probability	Impact	Risk Exposure EVM = Probability x Impact
Risk 1 (Threat)	10%	\$5000	\$500
Risk 2 (Threat)	90%	\$8000	\$7200
Risk 3 (Threat)	20%	\$4000	\$800
Risk 4 (Threat)	70%	\$2000	\$1400
Risk 5 (Opportunity)	80%	-\$1000	-\$800
Contingency Reserved Needed			\$9,100

Contingency Reserved Needed = [Sum of Threat] – [Sum of Opportunity]
= \$9,900 - \$800 = \$9100

Perform Quantitative Risk Analysis

Point to remember

- ✓ Here's a trick to remember the difference between qualitative and quantitative analyses: **Qualitative analysis** means that you're describing the qualities of the risks; quantitative analysis gives a quantity, usually a number, and often with a dollar sign next to it, of the impacts of those risks.

- ✓ **Qualitative analysis** is better for smaller projects (under $100,000 and shorter than three months). Larger projects, with more money at stake and longer durations, require more quantitative analysis.

PLAN RISK RESPONSE

This is where you decide whether to avoid, mitigate, transfer, or accept risk and how you'll do it.

Mitigate

In this type of risk response strategy, you try to minimize either the probability of the risks happening or the impact.

For example, you find that a team member from your team may leave for a certain duration during the peak of your project. Therefore, to minimize the impact of his absence, you identify another employee with similar qualifications from your organization and inform his boss that you may need him for your project for a period of time.

PLAN RISK RESPONSE

Transfer

In transfer risk response strategy, you transfer the risk to a third party to manage it. Please note that the transfer of risk does not eliminate the risk; it only transfers the responsibility of managing the risk to the third party.

For example, in your project there is a task to install some equipment and you don’t have much experience in this type of task. Therefore you ask a contractor to come and install it and sign a fixed price contract.

PLAN RISK RESPONSE

Avoid

Here you try to eliminate the risk or its impact on your project objective. You do this by either changing your project management plan, by making some changes to the project scope, or by changing the schedule.

For example, you observe that during certain periods of your project there is a chance of rain and you have some work planned outdoors at that time. Therefore, to avoid this risk, you move these activities to some other time to avoid the effect of rain.

Accept

This risk response strategy can be used with both kinds of risks, i.e. either positive risks or negative risks.

Risk Reduction Leverage

Risk reduction leverage = (RE_{before} - RE_{after})/ (Cost of Risk Reduction)

RE_{before} is risk exposure before risk reduction e.g. 1% chance of a fire causing £200k damage

RE_{after} is risk exposure after risk reduction e.g. fire alarm costing £500 reduces probability of fire damage to 0.5%

RRL = (1% of £200k)-(0.5% of £200k)/£500 = 2

RRL > 1.00 therefore worth doing

Difference between Qualitative and Quantitative Risk Analysis

The below table summarizes the difference between these two risk analysis.

Qualitative Risk Analysis	Quantitative Risk Analysis
Focuses on all the risks identified in the identify risk process.	Focuses on all the risks that have a possibility and high impact on the project elements.
Does not use numerical methods.	Uses mathematical calculations
Scales risks by using numbers (0-5) or percentages.	Calculates the effect of risk as a monetary value (cost) or number (duration).
It is applied to almost all projects	It is often applied to large and complex projects

Risk Response Plan

Risk ID	Name of Risk	Risk Response	Risk Generated	Risk Mitigation Actions
1	Wrong estimation	Accept	Delay in completing of the project	
2	Financial difficult by the contractor	Avoid	Delayed implementation of the commitments	
3	Financial difficult by the owner	Avoid	Delayed in disbursement of the advance	
4	Delay in completing of the project	Accept	Depression	
5	Change in cost of equipment	Avoid	Ability to construct	

SOFTWARE RISK MONITORING

Software risk monitoring is integrated into project activities and regular checks are conducted on top risks. Software risk monitoring comprises of:

- Implementing risk response plans
- Tracking identified risks
- Monitoring residual risks
- Identifying new risks
- Regularly search for new risks
- Evaluating risk process effectiveness throughout the project